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ANALYSIS OF 3D VSL GRAPHS WITH NEGATIVE VALUES OF VARIABLES

Last year, the Nobel Prize in Economics was awarded to three American economists who significantly improved the understanding of the role of banks in the economy, especially during financial crises, by giving money to banks and the population to restore purchasing activity in the country. Here it is worth recalling at once that in a number of countries, such as Sweden, Switzerland, Denmark and Japan, banks issued some customers with a negative rate, i.e. customers returned an amount less than they received [1, 2, 3]. Earlier, the author also published two articles about this problem [4, 5].

This article considers the issue of calculating the values of the lower critical volume of the Vsl spherical economic shell and constructing 3D drawings when the values of three variables change, i.e. $Vsl(GDPsl) = f(X1, X2, X3)$. In some cases, when the Vsl values had small minima and maxima compared to the last calculated Vsl value, then 3D drawings were built from the calculation so that the reader would understand more clearly how these changes occur.

Figure 1 below shows a 3D Vsl graph when the values of the variables were as follows $X1 = X2 = -1 \dots -10$, $X3 = 1$. Here the calculated Vsl parameter has no solutions. In the following Fig. 2 The depicted 3D Vsl graph for variables $X1 = X3 = 1$, $X2 = -1 \dots -10$ decreases from 0.64 to 0.02, i.e. 27.33 times.

If the variables are as follows $X1 = 1$, $X2 = -1 \dots -10$, $X3 = -1$, then in this case the Vsl value has no solutions, as can be seen from Fig. 3. In the following Figure 4, a curve was constructed at $X1 = X3 = -1$, $X2 = -1 \dots -10$. 4 and it shows that the Vsl parameter increases from 0.26 to 258.07, i.e. 1000 times.

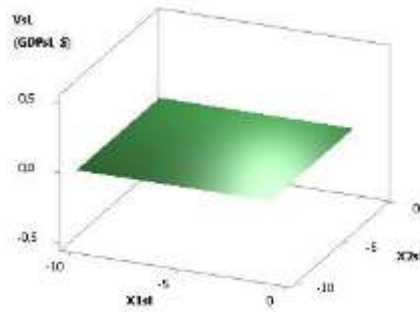


Fig. 1. 3D of Vsl (GDPsl)

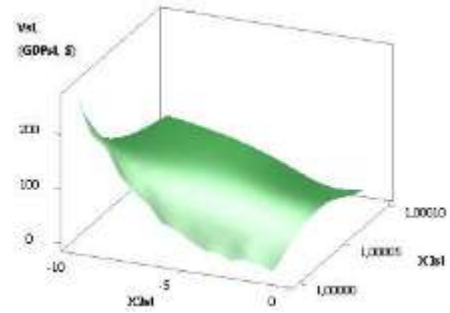


Fig. 2. 3D of Vsl (GDPsl)

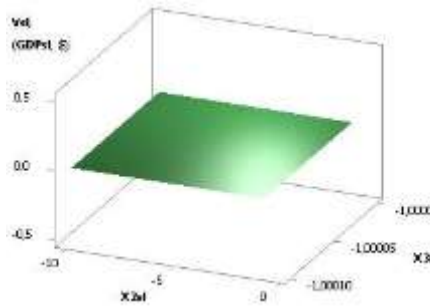


Fig. 3. 3D of Vsl (GDPsl)

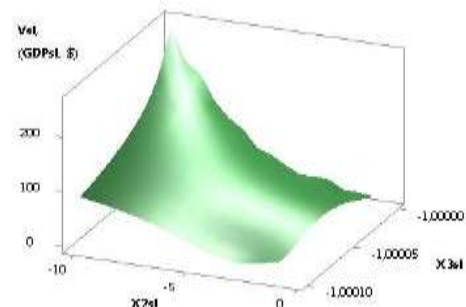


Fig. 4. 3D of Vsl (GDPsl)

The calculated values for the Vsl 3D graph in Figure 5 for variables $X1 = -1$, $X2 = 1$, $X3 = -10 \dots -1$ have no solutions. From the following Figure 6, it can be seen that for variables $X1 = X3 = -1 \dots -10$, $X2 = -1$, the Vsl values remain unchanged and equal 0.26.

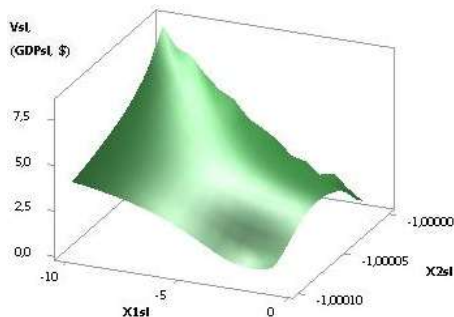


Fig. 5. 3D of Vsl (GDPsl)

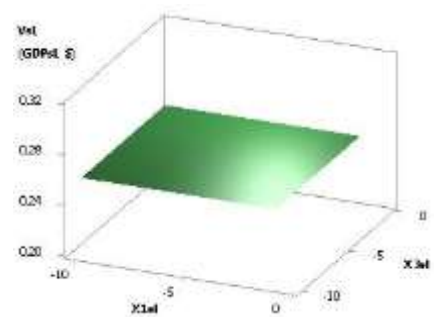


Fig. 6. 3D of Vsl (GDPsl)

Figures 7 and 8 were constructed at $X1 = X2 = -1$, $X3 = -10 \dots -1$ and $X1 = -1$, $X2 = X3 = -10 \dots -1$. Here, as in Figure 7, the Vsl values increase by 31.652 times, and in Figure 8 they decrease from 8.16 to 0.26, i.e. by 31.62 times.

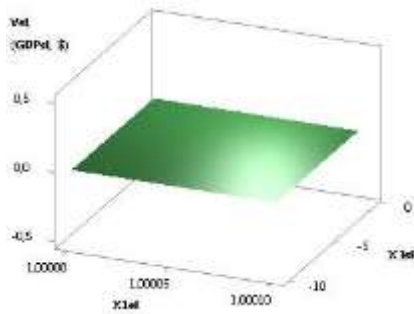


Fig. 7. 3D of Vsl (GDPsl)

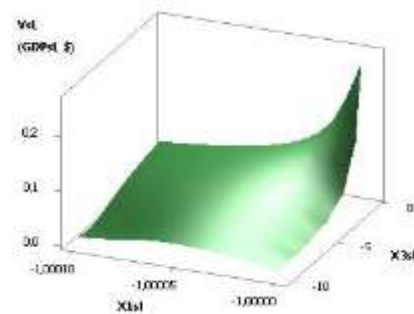


Fig. 8. 3D of Vsl (GDPsl)

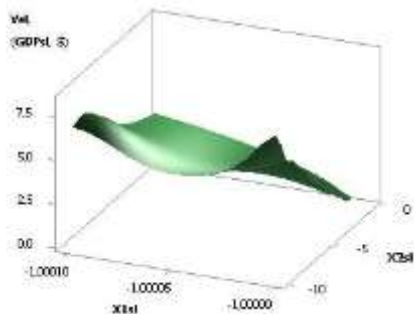


Fig. 9. 3D of Vsl (GDPsl)

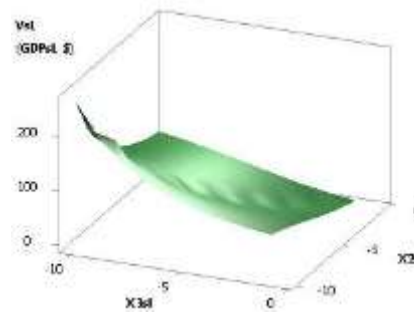


Fig. 10. 3D of Vsl (GDPsl)

In Figures 9 and 10 were constructed at $X1= X2= X3= -10...-1$ and $X1= -1, X2= 1, X3= -10...-1$. Here in Fig. 9, the Vsl parameter decreases significantly from 258.07 0 to 26, i.e. 1000 times, and in Fig. 10 it increases only 31.62 times.

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